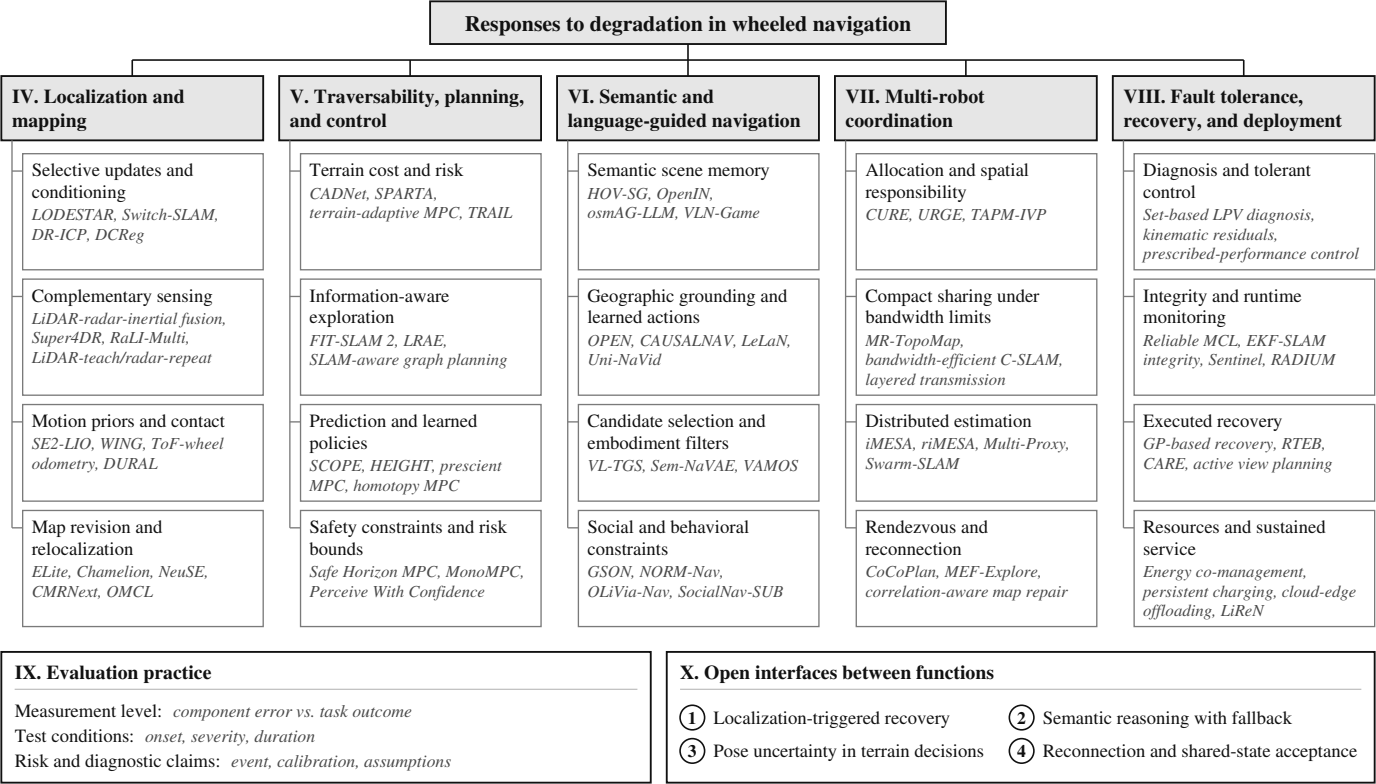

Responses to degradation in wheeled navigation
IV. Localization and mapping
Selective updates and conditioning
LODESTAR, Switch-SLAM, DR-ICP, DCReg
Complementary sensing
LiDAR-radar-inertial fusion, Super4DR, RaLI-Multi, LiDAR-teach/radar-repeat
Motion priors and contact
SE2-LIO, WING, ToF-wheel odometry, DURAL
Map revision and relocalization
ELite, Chamelion, NeuSE, CMRNext, OMCL
V. Traversability, planning, and control
Terrain cost and risk
CADNet, SPARTA, terrain-adaptive MPC, TRAIL
Information-aware exploration
FIT-SLAM 2, LRAE, SLAM-aware graph planning
Prediction and learned policies
SCOPE, HEIGHT, prescient MPC, homotopy MPC
Safety constraints and risk bounds
Safe Horizon MPC, MonoMPC, Perceive With Confidence
VI. Semantic and language-guided navigation
Semantic scene memory
HOV-SG, OpenIN, osmAG-LLM, VLN-Game
Geographic grounding and learned actions
OPEN, CAUSALNAV, LeLaN, Uni-NaVid
Candidate selection and embodiment filters
VL-TGS, Sem-NaVAE, VAMOS
Social and behavioral constraints
GSON, NORM-Nav, OLiVia-Nav, SocialNav-SUB
VII. Multi-robot coordination
Allocation and spatial responsibility
CURE, URGE, TAPM-IVP
Compact sharing under bandwidth limits
MR-TopoMap, bandwidth-efficient C-SLAM, layered transmission
Distributed estimation
iMESA, riMESA, Multi-Proxy, Swarm-SLAM
Rendezvous and reconnection
CoCoPlan, MEF-Explore, correlation-aware map repair
VIII. Fault tolerance, recovery, and deployment
Diagnosis and tolerant control
Set-based LPV diagnosis, kinematic residuals, prescribed-performance control
Integrity and runtime monitoring
Reliable MCL, EKF-SLAM integrity, Sentinel, RADIUM
Executed recovery
GP-based recovery, RTEB, CARE, active view planning
Resources and sustained service
Energy co-management, persistent charging, cloud-edge offloading, LiReN
IX. Evaluation practice
Measurement level: component error vs. task outcome
Test conditions: onset, severity, duration
Risk and diagnostic claims: event, calibration, assumptions
X. Open interfaces between functions
1 Localization-triggered recovery
2 Semantic reasoning with fallback
3 Pose uncertainty in terrain decisions
4 Reconnection and shared-state acceptance